Journal of Cleaner Production 108 (2015) 256–270

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Econological scheduling of a manufacturing enterprise operating under a time-of-use electricity tariff

Abhay Sharma [a, b, *], Fu Zhao [a, c], John W. Sutherland [c]

[a] School of Mechanical Engineering, Purdue University, West Lafayette, IN, USA
[b] Indian Institute of Technology Hyderabad, Hyderabad, India
[c] Division of Environmental and Ecological Engineering, Purdue University, West Lafayette, IN, USA

ARTICLE INFO

Article history:
Received 11 March 2014
Received in revised form
30 May 2015
Accepted 1 June 2015
Available online 10 June 2015

Keywords:
Sustainable manufacturing
Econological scheduling
Time-of-use tariff
Discrete event simulation

ABSTRACT

A new 'econological scheduling' model combining the economic and ecological aspects of a multi-part multi-machine setup operating under a time-of-use tariff is presented. The operating speed of the machines and the frequency of operating speed change are allowed to vary, and the peak load and energy consumption during a shift is estimated using discrete event simulation. The electricity cost and environmental impact for a target production quota are simultaneously minimized using a multi-criterion meta-heuristic optimization. The proposed model is demonstrated via a case study on a manufacturing unit producing parts using machining and welding operations. A comparison among econological, economic, and ecological approaches and the underlying dynamics of scheduling under a time-varying electricity tariff are presented as one of several strategies for enabling a manufacturing system to be more eco-friendly without substantially increasing the electricity cost.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Environmentally conscious manufacturing (ECM) has become more prevalent in today's world due to the significant increases in energy prices (Schlosser et al., 2011). The scope of the minimization of the environmental impact of manufacturing processes was initially limited to three principal methods: the modification of existing processes, replacement with alternative processes, and the development of new manufacturing processes (Munoz and Sheng, 1995). Subsequently, ECM was broadened to embrace a variety of other manufacturing issues, e.g., product design and development (Jackson et al., 1997) and systems planning (Sheng et al., 1998). The work of Munoz and Sheng (1995) and Dahmus and Gutowski (2004) addressed environmentally conscious machining. Basu and Sutherland (1999) introduced multi-variable multi-objective decision-making to simultaneously address cost, quality, and health hazard performance measures.

The current scope of the ECM is very broad and covers many different systems and scale levels, including unit processes, multiple machines, the factory, multiple facilities and even an entire supply chain (Duflou et al., 2012). Most of the work in the past focused on reducing the environmental impacts of manufacturing concentrated on smaller scales; for example, much of it was largely limited to unit processes or multi-machine approaches where a single product was the principal emphasis. The next highest level, that is, the factory level, involves multiple products that are simultaneously being produced using multiple machines. This scale invites a new challenge of reducing the non-productive time between operations. The non-productive time accounts for a significant portion of the cost and energy, as the utilities and supporting facilities continuously consume energy irrespective of production rate. As noted by Herrmann et al. (2011), the demand for higher productivity (cost, quality, and time) must be balanced with facility management and energy optimization because the cost of energy can be high enough to affect the plan for optimal facility operation.

Despite efforts to minimize peak loads and reduce the overall energy consumption within individual factories, power companies (electricity providers) are struggling to keep pace with growing electricity demands (this is especially true in the developing world).

As a consequence, energy shortages and peak demand deficits occur. Electric power companies apply different strategies for minimizing peak loads (Albadi and El-Saadany, 2008). Direct load control, interruptible load, and demand bidding are some of the

* Corresponding author. Indian Institute of Technology Hyderabad, Hyderabad, India.

http://dx.doi.org/10.1016/j.jclepro.2015.06.002
0959-6526/© 2015 Elsevier Ltd. All rights reserved.

incentive-based schemes in which energy saving provides a discount and overutilization a penalty. In contrast, price-based schemes are dynamic in the sense that the electricity rates change based on the real-time cost of electricity. These schemes are intended to smooth the demand pattern by charging a high price during peak periods and lower prices during off-peak periods. Time of use (TOU), extreme day pricing (EDP), and critical peak pricing (CPP) are some of the price-based schemes. Power companies around the world have introduced or are planning to introduce TOU tariffs to meet demand while balancing peak loads. A TOU tariff prescribes variable rates for electricity consumption during different times of day. A TOU tariff scheme, in principle, helps an electricity provider balance its load. The time-of-use tariff is based on the cost of producing the energy, which depends on the load seen by the electricity provider. The cost of production during a peak load period may be many times higher than that during off-peak periods; however, historically, power companies have not billed consumers based on the actual costs of production. A TOU tariff begins to align electricity charges with production costs. It encourages consumers to schedule their activities so that their electricity consumption (demand) is lower during the peak time. TOU tariffs are one of the most common demand-side management strategies, offering a means to lower electricity costs for consumers. However, a recent study by Jacopo (2012) showed that the under a TOU tariff, the average electricity consumption may sometimes increase, which translates into an increased carbon footprint.

A literature review suggests that no research has considered production scheduling based on electrical load tracking with the objective of the simultaneous minimization of the energy cost and carbon footprint for discrete manufacturing systems (DMSs). Dietrich (1991) provided a detailed discussion of classification, modeling, analysis, and optimization techniques for DMS. A DMS consists of many individual production machines/shops producing sub-parts or sub-assemblies, which are assembled to obtain the final product. The product can be disassembled at a later stage, and a sub-part may be replaced. The production of automobiles, electronic goods, and airplanes are examples of DMS. In contrast, continuous systems are concerned with the processing of raw materials and their blending into a final product in an irreversible fashion. Pharmaceuticals, fertilizers, and steel plants are examples of continuous systems. A continuous system of production is flexible and self-adjusting. Despite insufficient inventory, the production can continue with a smaller lot size (in terms of volume or weight), or a different formulation may be produced. This type of flexibility is not possible with DMS; the production needs to be halted in the absence of sufficient inventory required to assemble a product. Such halts in production are counterproductive and energy-consuming. Although no production takes place in this period, the basic amenities and utilities continue to consume energy. The conventional production scheduling practices have not addressed such issues. Thus, conscious efforts are essential for integrating the economic and ecologic (econologic) manufacturing scheduling models.

In the present work, we introduce a new ‘econological scheduling’ model for a discrete manufacturing system that aims to simultaneously optimize economic (energy cost) and ecological (carbon footprint) performance measures by the development of schedules that consider electrical load and the presence of a TOU tariff. The next section presents the background of the problem. This section is followed by the model formulation. Next, the proposed model is demonstrated using a case study on a manufacturing facility that uses a turning machine and a vertical machining center to produce sub-parts of a final product fabricated using a robotic welding machine.

2. Background

The scheduling problem in manufacturing has been widely investigated over the last 50 years. Extensive research has been conducted on various aspects of the scheduling problem, including job shop scheduling (e.g., Brucker et al., 1990; Gonçalves et al., 2005), flow shop scheduling (e.g., Osman and Potts, 1989; Tavakkoli at al. 2007), flexible job shop scheduling (e.g., Brandimarte, 1993; Pezzella et al., 2008), and open shop scheduling (e.g., Gonzalez and Sahni, 1976; Blum, C. 2005). Most of the research related to discrete manufacturing scheduling has adopted some simplifying assumptions. For example, although most real-life situations involve different types of unrelated machines, most of the modeling research has examined hybrid flow shop scheduling using identical machines (Marcus et al., 2011). Although some of the cases are related to real-life applications, the mathematical formulation is similar in the sense that either a single objective or mutually independent multiple objectives (usually only two objectives) have been considered. Efforts are continuously being made to solve scheduling problems using various techniques. For instance, over the last 10 years, meta-heuristic techniques such simulated annealing (Varadharajan and Rajendran, 2005), tabu search (Armentano and Arroyo, 2004), genetic optimization (Chang et al., 2007), particle swarm optimization (Chen, 2011), ant colony optimization (Leung et al., 2010), and a hybridization of the aforementioned techniques (Vallada et al., 2009; Zobolas, 2009; Sayadi et al., 2010; Mesmas et al., 2011; Yin-Yann et al., 2013) have been used to address scheduling problems.

Despite these enormous efforts, the outcomes of the scheduling investigations have largely centered on establishing a better algorithm that can yield a superior solution to the best known solution. In retrospect, considering the interdependencies of the machines, building, and facilities and associated complexities, simulation and optimization might be better applied to achieving the best operating environment and production plan. With such an approach, scheduling simulation would become an integral part of ECM at the factory level. Limited attempts relating to energy-conscious scheduling have been made in the past (Swaminathan and Chakrabarty, 2003; Subai and Baptiste, 2006; Mouzon and Yildirim, 2008; Róbert et al., 2008; Maria et al., 2008; He at al. 2008, He at al. 2012, Wang and Huang, 2009; Natasha et al., 2010; Yoon et al., 2013; Shrouf at al. 2014). Fang et al. (2011) presented a multi-machine level multi-objective scheduling simulation aimed at the simultaneous optimization of the makespan, the peak power load, and the carbon footprint for the first time. Unlike conventional scheduling aimed at maximizing machine/operation speed to maximize productivity, the formulation presented by Fang et al. (2011) allows the operational speed to vary to minimize the peak load and energy consumption, in addition to maximizing productivity. More recently, Fadi et al. (2014) proposed a mathematical model to minimize energy consumption costs for single-machine production scheduling by making decisions about job launch time, idle time, turning-on time, and turning-off time. Sun and Li (2014) presented an analytical model for the identification of the optimal energy control actions through changes in the system state of an assembly line.

It should be noted that recent efforts with the goal of minimizing the total operating cost and satisfying production targets through peak electrical load adjustment while operating under a TOU tariff (Ashok, 2006; Kristian and Manfred, 2010; Alain and Christian, 2011) have concentrated on scheduling continuous production systems, e.g., steel plants and electrolytic process systems.

A DMS that has multiple machines and processes multiple parts is most likely to benefit from electrical load tracking similar to that applied to continuous production systems; however, it is believed that the former is a more challenging technical task. Unlike a continuous system, attributes such as the dependency of job completion time on part geometry, the wait for the availability of a machine for the next operation, and line balancing, etc. make the scheduling of a DMS more complicated. These types of challenges do not exist in continuous production systems. Moreover, to change the peak power demand, according to a load management strategy, the operating speeds of the machines need to be altered. However, the speeds of all of the machines cannot be altered simultaneously, as such an action will cause jobs on various machines to complete at different times. An additional speed-changing algorithm needs to be developed and coupled with the scheduling problem. The size of the shop, i.e., the number of parts and machines, is another challenge. With an increment in the shop size, it becomes more difficult computationally to find a scheduling solution for multi-criteria optimization.

Among the aforementioned challenges in the scheduling of DMS, dependency of job completion time on part geometry, the wait for the availability of a machine for the next operation, change in peak power demand by an operating-speed-changing algorithm, and size of the shop have been addressed in this paper. A speed-changing algorithm is coupled with a scheduling algorithm. The peak load and energy consumption are tracked using discrete event simulation. An optimum schedule with minimum electricity cost and minimum carbon footprint operating under a TOU tariff is achieved through meta-heuristic optimization. In the next section, the proposed econological scheduling model is presented. A case study is then employed to demonstrate the proposed model.

3. Problem description

3.1. Manufacturing set-up

The manufacturing set-up consists of $M = \{M_1, M_2, \ldots M_i, \ldots \ldots, M_m\}$ machines. Each machine can run at a finite and discrete set of operating speeds $S_i = \{s_1, s_2, \ldots s_h, \ldots \ldots, s_i\}$. The machines produce different sub-parts, partially or fully, as per the design. Each sub-part is processed on one or more machines. The sub-parts are eventually assembled to produce a part. All of the machines are allowed k_i number of times each day (or a stipulated time duration, e.g. an eight hour shift) when their operating speed may be changed. No speed consistency among the machines is considered, i.e., the speeds are independently selected for each machine. Thus, at any instant, the operating speed of a machine is $s_{ik} \in S_i$. To add to the complexity, sub-parts, each in different numbers, contribute to the final assembly, and all of the sub-parts do not follow the same machining operations and sequence. The operational speeds of different machines are allowed to vary to match the production target and the energy consumption. The choice of speed is affected by the time of operation, as the enterprise runs under a time-dependent TOU tariff. Because one machine is available for a type of manufacturing operation, viz. line balancing is not considered. The failure of a machine and maintenance are not considered.

3.2. Electrical tariff

The electricity charge (E_c) under a TOU tariff varies throughout the day. In fact, the tariff is most often characterized as a discrete function, with the charges depending on the time period (block) during the day. For a given block, two charges are taken into account, namely, the electricity charge and the demand charge. The electricity charge is calculated based on the consumed electrical energy during the block, and the demand charge is calculated based on the maximum usage, i.e., the peak load during the block.

A general model of the electricity charge for a day under a TOU tariff is

$$E_c = C_0 + \sum_{g=1}^{b} (C_{1g} P_g + C_{2g} E_g) \quad (1)$$

where $g = 1, \ldots, b$ is the number of time blocks per day, C_0 is the fixed cost per day, C_{1g} is the coefficient of demand (the cost per unit peak load (cost/kW)), P_g is the peak demand in kW (peak load) during the gth period, C_{2g} is the coefficient of energy (cost per unit energy (cost/kWh)), and E_g is the energy consumed in kWh for the gth period.

3.3. Speed change model

To minimize the electricity cost, the peak load and the energy consumption of the enterprise need to be synchronized with the TOU block charges. Changing the operating speeds of the machines is perhaps the most straightforward method to change the peak load and energy consumption. A speed change also affects the completion time of an individual sub-part, which in turn affects the waiting time of the sub-part before the next operation starts. There are at least two options for changing speeds. In option 1, for every sub-part, the machine is allowed to run at a user-specified speed. In option 2, the operating speeds of the machines are allowed to change at regular intervals such that in an instant, every machine runs independently on any of the possible operating speeds. The former gives more flexibility in choosing the speed-schedule to achieve a given production target; however, this approach may become excessively complex (and thus prohibitive in terms of computation time) when the number of machines or parts is excessively high. In the present work, the latter option is used, i.e., all of the machines are allowed a k_i number of times each day when their operating speed may be changed. Thus, at any instant, the operating speed of a machine is $s_{ik} \in S_i$. If a sub-part is not finished processing on a machine at the time of a designated speed change, the speed is adjusted when the operation on the sub-part is completed.

3.4. Operating variables

A preliminary study on electricity cost and energy consumption with random speed-schedules and a varying number of parts and number of speed changes in a shift of eight hours (8:00 am to 4:00 pm) is conducted as a part of this investigation to understand the econological dynamics of the DMS. Fig. 1(a) and (b) show the effect of the number of parts (n_p) on the electricity cost per part and total energy consumption. Fig. 1(a) shows that the electricity cost per part reduced by limiting n_p (using lower speeds); however, Fig. 1(b) shows that such a change may demand a higher energy consumption. This result is undoubtedly because the basic and idle components of the electrical load will take on increasing importance as the speeds are lowered to reduce processing power. This is an important outcome indicating that the operating speeds, and in turn the amount of production, must be meticulously managed to obtain the maximum economic and ecological benefits.

Fig. 1(c) and (d) show the effect of the number of speed changes on the electricity cost per part and total energy consumption. As the figures show, when the number of speed changes is small, the machines cannot adjust their operating speeds and respond to the electrical tariff. As a result, a small number of speed changes generally produce both higher electricity cost per part and

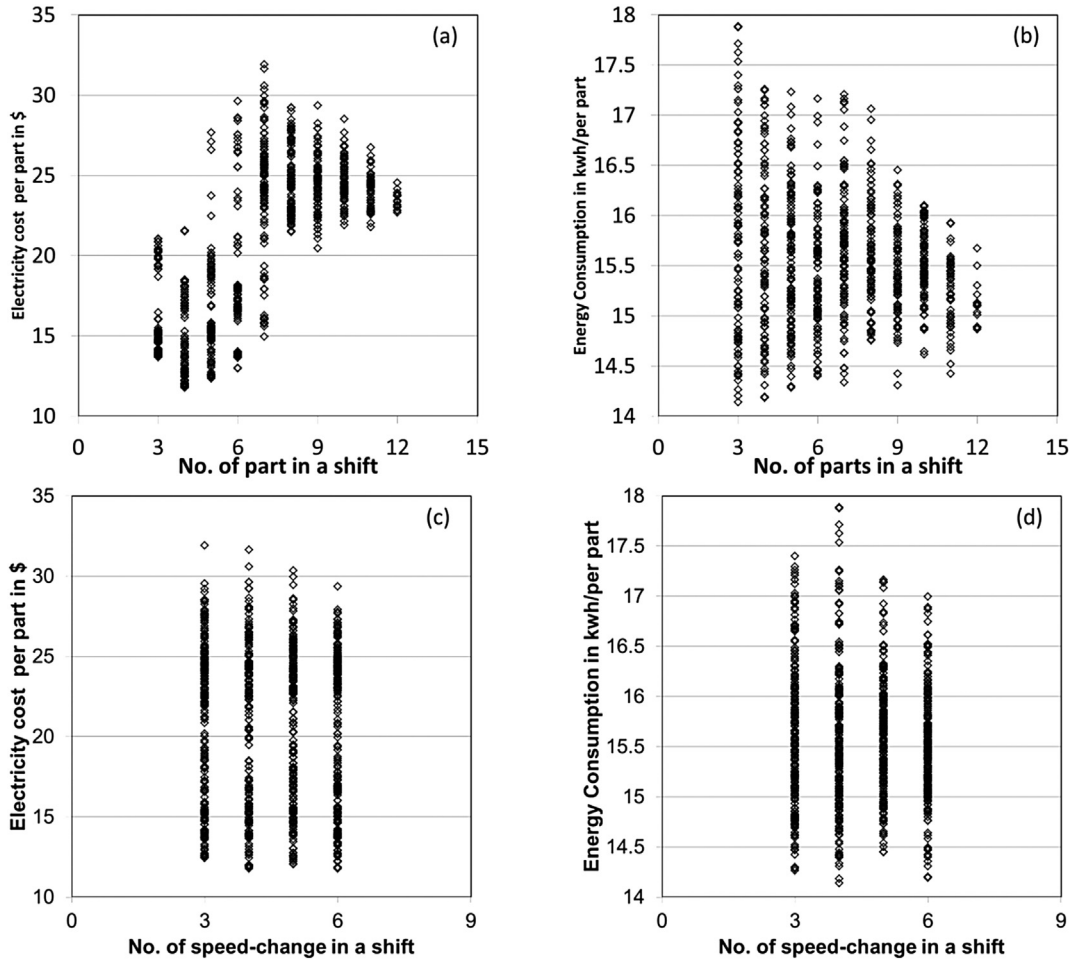


Fig. 1. Effect of (a) the number of parts in a shift on electricity cost, (b) the number of parts in a shift on energy consumption, (c) the number of speed changes in a shift on electricity cost, and (d) the number of speed changes in a shift on energy consumption.

increased electricity (energy) consumption. Based on the foregoing discussion, the operating speed (s_{ik}) and the number of permissible speed changes (k_i) are considered to be decision variables in the econological optimization of the DSM scheduling problem.

3.5. Load tracking

The electrical load (power) is tracked using discrete event simulation. The process flow of each sub-part, provided in Appendix 1 for the test-case described shortly, gives the sequence of activities and the associated processing times. The electrical load on each machine is recorded at regular time intervals. The instantaneous load at a given point of time can be calculated based on the following equation:

$$P = \sum_{i=1}^{m} (P_i)_{basic} + (P_i)_{idle} + (P_i)_{operation} \quad (2)$$

The maximum value of P in the gth period gives the peak demand P_g. The subscript 'basic' refers to such activities as workpiece loading/unloading, positioning, and fixturing. The activities associated with the utilities and auxiliary units also count for the basic activities, however, in the present investigation, the basic activities of production machines are only considered. The subscript 'idle' refers to such activities as tool approaching and retracting from the work-piece, moving tools between features, adjusting machine settings, and changing the tools. The subscript 'operation' refers to the actual processing, such as cutting and welding operation.

Due to the different operating speeds of the various machines, the typical load for each machine, and therefore the total electrical (power) load, may vary for the multi-machine setup as shown in Fig. 2. The area under the curve for a time period gives the electricity (energy) consumption for that time period. To calculate the electrical load, each period is divided into small time intervals (Δt). The operating speed at the time midpoint of the interval is used to calculate the electrical load. The Δt played an important role in the accurate calculation of the energy consumption and the simulation time. Beyond a certain limit, further narrowing the time interval did not considerably affect the accuracy of the calculations of load and energy; however, the computation time increased drastically. A time interval of one minute gave a reasonable trade-off between accuracy and computation time.

3.6. Mathematical model

The manufacturing setup described earlier is a two-stage assembly flow shop scheduling problem initially appeared in work of Lee et al. (1993) and Potts et al. (1995). The first stage consists of multiple concurrent operations which are followed by a single

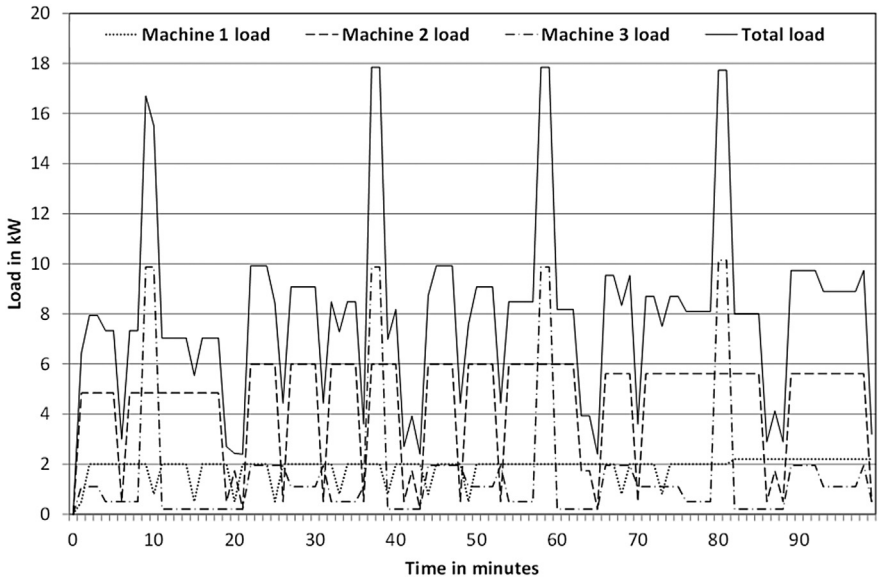


Fig. 2. Typical electrical load pattern for a multi-machine DMS.

assembly operation in the second stage. The two stage assembly flow shop is defined as follows:

The scheduling environment consists of two stages; $M = \{M_1, M_2, \ldots M_j, \ldots \ldots, M_m\}$ machines at the first stage and one machine at the second stage. There are n jobs to be scheduled such that total $m + 1$ operations (i.e. m concurrently in the first stage and one in the second stage) are performed. The first stage operations are performed by different machines and each machine can perform only one operation at a time. The second stage operation start only when all m operations of the first stage are completed. All jobs are available at time zero and no preemption is allowed. The two-stage assembly problem may also be considered as m artificial two-machines flow-shops wherein ith flow-shop consists of machines M_i of the first stage and machine M_{m+1} of assembly stage (Potts et al., 1995), as shown in Fig. 3 (a).

Let

$t_{i,j}$: processing time of job i on machine j (at stage one), $i = 1, \ldots \ldots, n, j = 1, \ldots \ldots, m$
a_i: operation time of job i at stage two
C_i: completion time of the job i

It has been shown by Potts et al. (1995) that permutation schedules are dominant with respect to makespan, i.e., the sequence of jobs on all $m + 1$ machines remains same. If the given position of the job is denoted by the bracket, as shown below, the completion time of the job in position j is given as follows (Al-Anzi and Allahverdi, 2009):

$$C_{[j]} = max\left\{ \max_{k=1,\ldots\ldots,m} \left\{ \sum_{i=1}^{j} t_{[i,k]} \right\}, C_{[i-1]} \right\} + a_{[j]} \quad (3)$$

where $C_{[0]} = 0$ and makespan (C_{max}) is given as follows:

$$C_{max} = C_{[n]} \quad (4)$$

The present investigation considers a more complex situation wherein the processing times on machines are not known in advance since machines are allowed k_i times to change operating speed in stipulated time duration, e.g. a shift of eight hours, as discussed earlier. Each machine can run at a finite and discrete set of operating speeds $S_i = \{s_1, s_2, \ldots s_h, \ldots \ldots, s_i\}$. The present scheduling model is an extension of conventional two-stage assembly flow-shop model (Fig. 3(a)). The operating speed and number of speed changes are fed to the shop flow model and part schedule and speed schedule are calculated, as shown in Fig. 3(b). The optimum part schedule and speed schedule are obtained through heuristic optimization. The feasible schedule is given by assignment of jobs and operating speeds that can at least complete number of target jobs (n_t) with permissible number of speed changes in stipulated time duration. Three types of objective functions economical, ecological, and econological are considered. The economical objective is to minimize the electricity cost and maximize number of jobs in a shift whereas the ecological objective is to minimize the electricity consumption. The econological objective combines economical and ecological objectives and aims to minimize the electricity cost and electricity consumption, and maximize number of jobs. The mathematical formulation of objective functions is as follows:

Consider

P is instantaneous power consumption at any point of time because of M machines, as given in Eq. (2)
P_g is the peak demand in gth time block, obtained from load tracking as shown in Fig. 2
E_g is the energy consumed in gth time block; the area under the load tracking curve gives energy consumption
E_c is the total energy consumed in shift of total b blocks, as given in Eq. (1)
n_t is number of target jobs in a shift
n_p is number of actual jobs completed in a shift, $n_p \geq n_t$
C_f is the carbon footprint due to the production of n_p jobs $= \alpha \sum_{g=1}^{b} E_g$, where α is the carbon footprint per unit of electricity consumed (kg CO_2 eq/kWh).

Let the electricity cost per part, carbon footprint per jobs and number of jobs are expressed in non-dimensional sub-indices as O_e, O_f, and O_p, respectively, and given as follows:

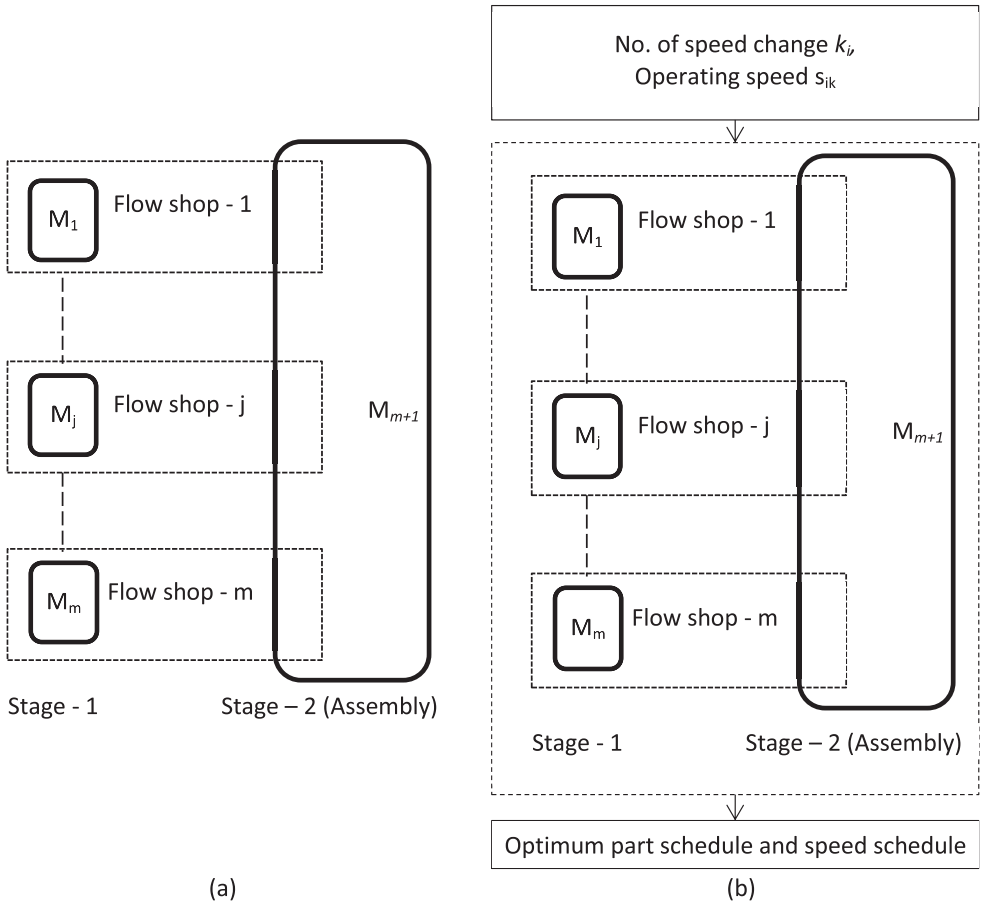


Fig. 3. Two-stage assembly flow shop (a) conventional (b) present investigation.

$$O_e = \left(\frac{\frac{E_c}{n_p} - \left(\frac{E_c}{n_p}\right)^{min}}{\left(\frac{E_c}{n_p}\right)^{max} - \left(\frac{E_c}{n_p}\right)^{min}} \right) \tag{5a}$$

$$O_f = \left(\frac{\frac{C_f}{n_p} - \left(\frac{C_f}{n_p}\right)^{min}}{\left(\frac{C_f}{n_p}\right)^{max} - \left(\frac{C_f}{n_p}\right)^{min}} \right) \tag{5b}$$

$$O_p = \left(\frac{n_p^{max} - n_p}{n_p^{max} - n_p^{min}} \right) \tag{5c}$$

The sub-indices are defined considering minimization of objective functions such that the least value of sub-indices represents the expectation form the objective, e.g. when the number of actual jobs completed in a shift reaches maximum possible value, sub-index O_p tends to zero.

The economical, ecological, and econological objective functions are defined as follows:

$$\text{Economical} \quad \begin{matrix} minimize\ O_e \\ minimize\ O_p \end{matrix} \tag{6}$$

$$\text{Ecological} \quad minimize\ O_f \tag{7}$$

$$\text{Econological} \quad \begin{matrix} minimize\ O_e \\ minimize\ O_p \\ minimize\ O_f \end{matrix} \tag{8}$$

$$\text{subject to} \quad n_p \geq nt \tag{9a}$$

$$s_{ik} \in S_i = \{s_1, s_2, s_3 \ldots\ldots s_h \ldots\ldots s_i\} \tag{9b}$$

$$k_i \leq k_{max} \tag{9c}$$

The minimum and maximum values of the electricity cost per job, the carbon footprint per job, and the number of jobs per shift $[(E_C/n_p)^{min}, (E_C/n_p)^{max}, (C_f/n_p)^{min}, (C_f/n_p)^{max}, n_p^{min}, n_p^{max}$, respectively) are needed to create the non-dimensional sub-indices of Eq. (5a,b,c). These values are not known a priori, as they depend on the part geometry, production schedule, and selected machine speeds. Because the proposed model needs these values to function, a strategy is needed to estimate them. To estimate the maximum number of jobs per shift, all machines are set at their maximum

speed, and it is assumed there are no wait times for a job to be processed on a given machine. To find the minimum number of jobs per shift, all of the machines are set at their lowest speed. The minimum and maximum carbon footprint per part is estimated by simulating production of a job considering all of the possible speed combinations on all machines using. The minimum and maximum electricity cost per job are estimated by simulating the production of varying numbers of jobs under large number of random schedules with different numbers of speed changes. The decision variables, i.e., the operational speed (s_{ik}) and number of speed changes (k_i), are allowed to vary, and meta-heuristic optimization technique, i.e. simulated annealing is used to obtain a set of decision variables that minimizes the objective functions, as presented in the next section.

3.7. Solution scheme

Multi-objective scheduling optimization presents a significant technical challenge. As observed by Fang et al. (2011), even when scheduling a moderate number of parts, it was almost time-prohibitive to compute even one point on the Pareto frontier with multi-objective optimization. In general, if multiple objectives are to be optimized concurrently, designing a search algorithm is difficult, and its implementation is complex. One strategy for dealing with multiple objectives is to combine them into a scalar function using weights; in this way, a feasible solution can be found within a reasonable time. However, developing an appropriate and defendable set of weights is not an easy task. In the present paper, we hybridized two techniques, aggregate indexing and simulated annealing, to address the foregoing issues.

3.7.1. The aggregate quality index method

The commonly used methods for combining multiple objectives in a scalar function for optimization problems suffer from two complications, eclipsing and ambiguity (Prabhat and Aditya, 2000). Eclipsing is a condition in which the scalar function does not sense the weaker objectives (numerically smaller contributions to the overall objective function) because one or more strong objectives dominate the combined objective. The commonly used weighted-sum approach is one of the methods that exhibits such an eclipsing condition. Ambiguity is a condition where an overall acceptable set of conditions is pronounced as unacceptable by the scalar function. The harmonic mean is one objective that can demonstrate such ambiguity. Based on their study of aggregation, Prabhat and Aditya (2000) proposed the following aggregate index that is free of eclipsing and ambiguity:

$$I = \left(1 - z + \sum_{1}^{z} O_i^{\frac{-1}{\beta}}\right)^{-\beta} \tag{10}$$

where β is a positive constant, z is the number of objectives, and O_i is the ith sub-index (because I is an index, O_i may be termed a sub-index) expressed in non-dimensional form (between 0 and 1). It has been reported by Prabhat and Aditya (2000) that $\beta = 0.4$ provides a satisfactory aggregation of the objectives. In addition to the foregoing advantage of converting multiple objectives into a single objective without eclipsing or ambiguity, the aggregate index of Eq. (10) has some additional significant merits. The aggregate index is characteristically biased towards the weakest objective, i.e., it does not allow the aggregate index to exceed the minimum sub-index. However, for the same minimum sub-indices of the two sets of objectives, set 1 and set 2, the aggregate index can differentiate between the two sets on the basis of other sub-indices, e.g., the minimum sub-index of set 1 = the minimum sub-index of set 2 and the maximum sub-index of set 1> the minimum sub-index of set 2 leads to $I_1 > I_2$. This type of stringent index is essential to prevent the eclipsing of the ecological or economic concerns and, at the same time, to find the best of the two. Furthermore, the index being recursive, as shown in Eq. (11), allows the addition of an objective without repeating the aggregation exercise of Eq. (10).

$$I_{n+1}^{\frac{-1}{\beta}} = I_n^{\frac{-1}{\beta}} + O_{n+1}^{\frac{-1}{\beta}} - 1 \tag{11}$$

3.7.2. Simulated annealing

Simulated annealing is a systemic search-based meta-heuristic method that is considered very effective and less time-consuming than other methods, such as genetic algorithms. In simulated annealing, a series of randomly generated feasible solution vectors is generated in successive iterations. The flow chart of Fig. 4 shows the simulated annealing algorithm.

Each iteration is signified by a parameter called temperature. The algorithm starts with an initial temperature T_s and initial solution $[C]^0$. A small, randomly generated perturbation is applied to this solution, and the change in the objective function f is calculated. For the minimization problem, if $\Delta f \leq 0$, the perturbed solution is accepted; otherwise, the perturbed solution is accepted with a probability of

$$P(\Delta f) = e^{-\frac{\Delta f}{\gamma T}} \tag{12}$$

where γ is a scaling factor (Boltzmann constant) and T is the temperature at a given iteration. When a new solution is accepted, the temperature T is reduced by a decaying ratio C_T. This procedure is repeated until the final temperature T_f is reached, at which point no further improvement in the objective function can be expected. Simulated annealing is chosen because of the limitation of conventional approaches in solving large multi-objective scheduling problems. Fang et al. (2011) used integer programming for a multi-machine-level multi-objective scheduling

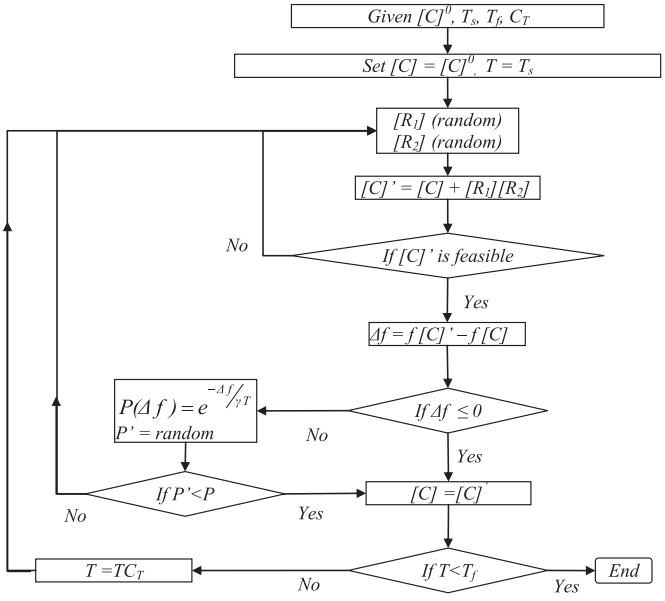


Fig. 4. Simulated annealing algorithm.

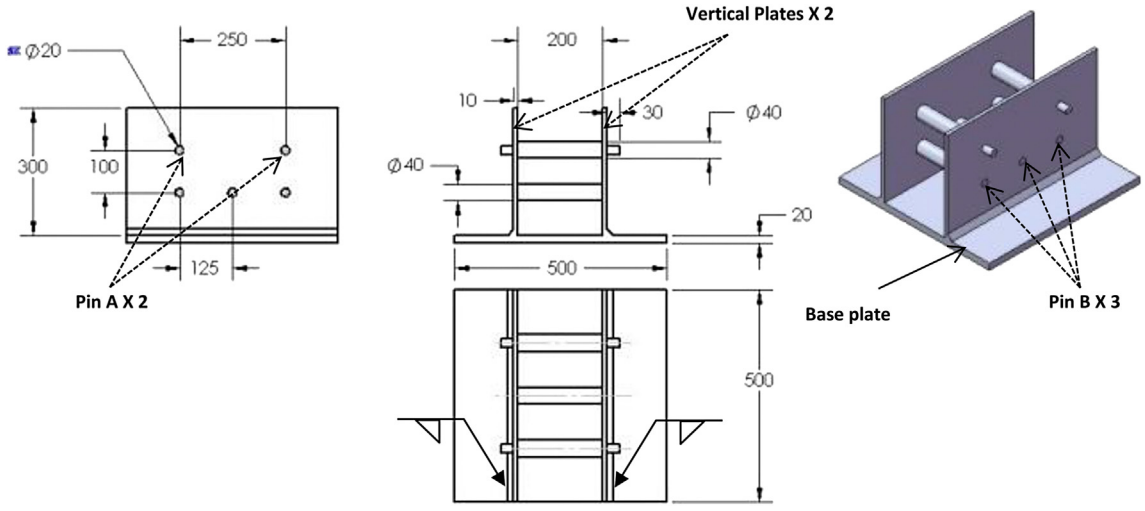


Fig. 5. Part geometry.

Table 1
Machining process conditions.

Parameter	Turning ($i = 1$)	Milling ($i = 2$)	Drilling ($i = 3$)
s_{ik} (m/min) – level 1/2/3/4/5	18/24/30/36/42	18/24/30/36/42	16/17/18/19/20
f_r (mm/rev)	0.2	0.2	0.2
d mm	1 mm	2	–

problem, similar to the one addressed in the present investigation. They observed that for a moderate number of jobs, e.g., 36, it was almost time-prohibitive to solve the problem using commercial optimization software tools.

4. Case study

The proposed model is demonstrated through a case study focused on manufacturing the part shown in Fig. 5. The final product considered in the present investigation is an assembly of many sub-parts, namely, pins (A and B) and plates (base plate and vertical plates). For practical reasons, henceforth, the completed job, i.e. job processed at machine M_m, is denoted as part and rest as sub-parts.

The individual sub-parts are produced using a turning machine and vertical machining center and are then assembled and welded at the robotic welding station following the operating conditions and the electricity tariff provided in Tables 1–3. Turning, milling, drilling, and welding operations are used. These operations during the kth speed changeover are performed at operational speed at s_{1k}, s_{2k}, s_{3k}, and s_{4k}, respectively. The following relations are used to calculate the power consumption and time associated with the various operations:

$$t_{turning} = \frac{\pi LD}{f_r s_{1k}} \tag{13}$$

Table 2
Welding process conditions.

Parameter	Welding ($i = 4$)
V (V)	30
Q (A)	290
s_{ik} (m/min) – level 1/2/3/4/5	0.34/0.35/0.36/0.37/0.38
P_t (kW)	1

$$P_{turning} = u \cdot f_r \cdot s_{1k} \cdot d \tag{14}$$

$$t_{milling} = \frac{\pi . D_c (L + D_c)}{n_e f_t \cdot s_{2k}} \tag{15}$$

$$P_{milling} = \frac{u \cdot W \cdot f_t \cdot s_{2k} \cdot d}{\pi} \tag{16}$$

$$t_{drilling} = \frac{L \pi D_c}{n_e f_t s_{3k}} \tag{17}$$

$$P_{drilling} = \frac{u \cdot n_e f_t \cdot s_{3k} \cdot D_c}{4} \tag{18}$$

$$t_{welding} = \frac{L}{s_{4k}} \tag{19}$$

$$P_{welding} = VQ + P_t \tag{20}$$

where d is the depth of the cut, D is the workpiece diameter in turning, D_c is the cutter diameter, f_r is the feed per revolution, f_t is the feed per cutting edge, n_e is the number of cutting edges, W is the width of the cut, Q is the welding current, L is the length of the cut (or weld), u is the specific cutting energy, V is the welding voltage, and P_t is the power consumed in the welding torch movement.

The process flow for the production of pins, base plates, and vertical plates and their assembly is explained in Appendix 1. The time and power consumption for idle and basic activities are predetermined, whereas Equations 13–20 are used to calculate the power and processing time for operational activities, which are then used to find the carbon footprint, electricity cost, and number of parts produced for a given schedule. The calculation of power and processing time is explained in Appendix 2. These quantities

Table 3
Electrical tariff.

Meter charge	USD 11.14 per day				
Time of use (blocks)	Monday through Friday (except holidays)				Saturday, Sunday, and Holidays
	Off-peak	Part-peak	Peak	Part-peak	
	9:30 pm–8:30 am	8:30 am–12:00 noon	12:00 noon–6:00 pm	6:00 pm–9:30 pm	All day
Demand Charge	–	4.25 USD/kW	13.30 USD/kW	4.25 USD/kW	–
Energy Charge	0.0753 USD/kWh	0.0995 USD/kWh	0.1356 USD/kWh	0.0995 USD/kWh	0.0753 USD/kWh

are then used to calculate the aggregate index. The aggregate indices for economical (I_x), ecological (I_y), and econological (I_{xy}) perspectives are defined as

$$I_x = \left(O_e^{-2.5} + O_p^{-2.5} - 1\right)^{-0.4} \tag{21}$$

$$I_y = O_f \tag{22}$$

Inserting Eqs. (21) and (22) into Eq. (11) produces the following relation:

$$I_{xy}^{-2.5} = I_x^{-2.5} + I_y^{-2.5} - 1 \tag{23}$$

which simplifies to

$$I_{xy} = \left(O_e^{-2.5} + O_f^{-2.5} + O_p^{-2.5} - 2\right)^{-0.4} \tag{24}$$

The (minimum–maximum) limits obtained for the number of parts per shift, energy per part, and electricity cost per part are 3–15, 14–18 kWh, and USD10–USD35, respectively.

Algorithms for the speed changes and scheduling are executed through a code written in MATLAB. The code was tested with different initial speed-schedules (s_{ik}) generated using the built-in random integer number generator 'randi'. Random integers between 1 and 5 were generated and each random integer corresponded to the five speed level of machines shown in Tables 1 and 2. Thus, the neighborhood structure of the problem consist of 4×5 (4 machines and 5 speed levels) matrix of random integers between 1 and 5. The Boltzman acceptance function (as shown in Eq. (12)) was used with values for the initial temperature, the final temperature, and the Boltzmann constant set at 100, 1×10^{-7}, and 1×10^{4} respectively. For a 4 machine set up with maximum allowable speed changeovers set at 10, it took 8 h on a computer with Intel® Core™ i5-4200M (3.10 GHz 400 MHz 3 MB) processors and 8 GB of RAM. The errors in the optimum power and energy values are limited to 0.04%. The optimization exercise is conducted from different perspectives to understand the underlying dynamics of energy, cost, and tariff. Some of the important observations are provided below.

4.1. Econological vs. economic and ecological

To gauge the effectiveness of the econological model, scheduling optimization is carried out for a case in which the target production is nine parts on a shift. Table 4 shows the result of this exercise *vis-à-vis* the cost, energy, and makespan for the three approaches obtained for the production target of nine parts in a shift. It can be observed that the ecological optimization produces a higher cost of electricity and makespan than the other two objectives. The sole objective of energy minimization leads to a reduction of the speed, resulting in a lower peak load, which eventually results in a larger makespan. Although the total energy consumption decreases due to the lower speeds, the production cycle spreads more into the peak period, resulting in higher cost. On the other hand, the economical optimization yields the minimum cost but at the expense of requiring more energy.

The foregoing facts can be verified through examination of the load spectrum obtained for the three approaches, as shown in Fig. 6. It can be observed that the ecological schedule consistently yields the minimum peak loads among the three approaches. On the other hand, the economical schedule most often results in the maximum peak load. It is interesting to note from the same figure that the econological schedule yields higher peak loads, similar to the economical schedule, in the first half of the shift, which consists of off-peak and part-peak electricity tariffs. As soon as the tariff changes to peak rate in the second half (after 12:00 pm), the peak loads of the econological schedule decrease to the same level as those of the ecological schedule.

The econological approach combines the benefits of the other two objectives and is able to find a schedule that can produce parts at the same electricity cost and energy consumption as the economic and ecological approaches do. The importance of the aggregate index in obtaining these results cannot be denied; these results may not have been obtained using conventional methods for combining objectives. The recursive character of the aggregate index ensures that individual objectives are considered. However, the econological approach needs to be examined against varying production targets, as will be described in the next section.

Table 4
Comparison of the ecological, economical, and econological approaches.

	Energy (kWh/part)	Electricity cost per part (USD)	Makespan (min)
Ecological	14.64	23.30	331
Economical	14.71	21.95	325
Econological	14.65	21.96	323
minimum value			

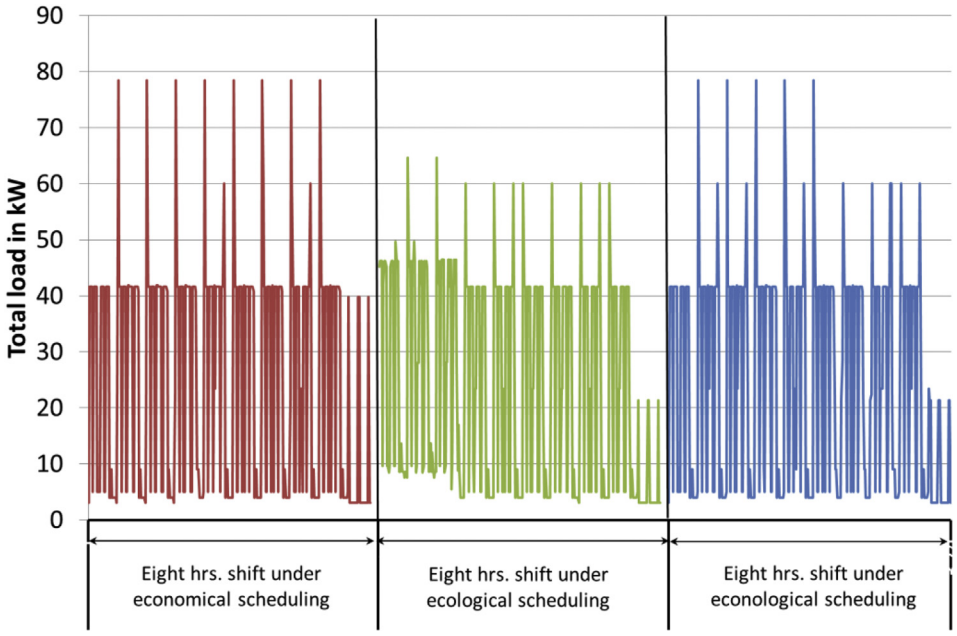


Fig. 6. Load spectrum for various scheduling approaches.

4.2. Econological vs. production target

The benefits of econological scheduling are production-target-dependent. The constraint due to the production target forces the machine(s) to run at higher speed(s). However, the scheduling counterbalances to an extent the effects of higher speeds at one or more machines by mutually adjusting the speeds at other machines. The same can be observed in Fig. 7, which shows the speed

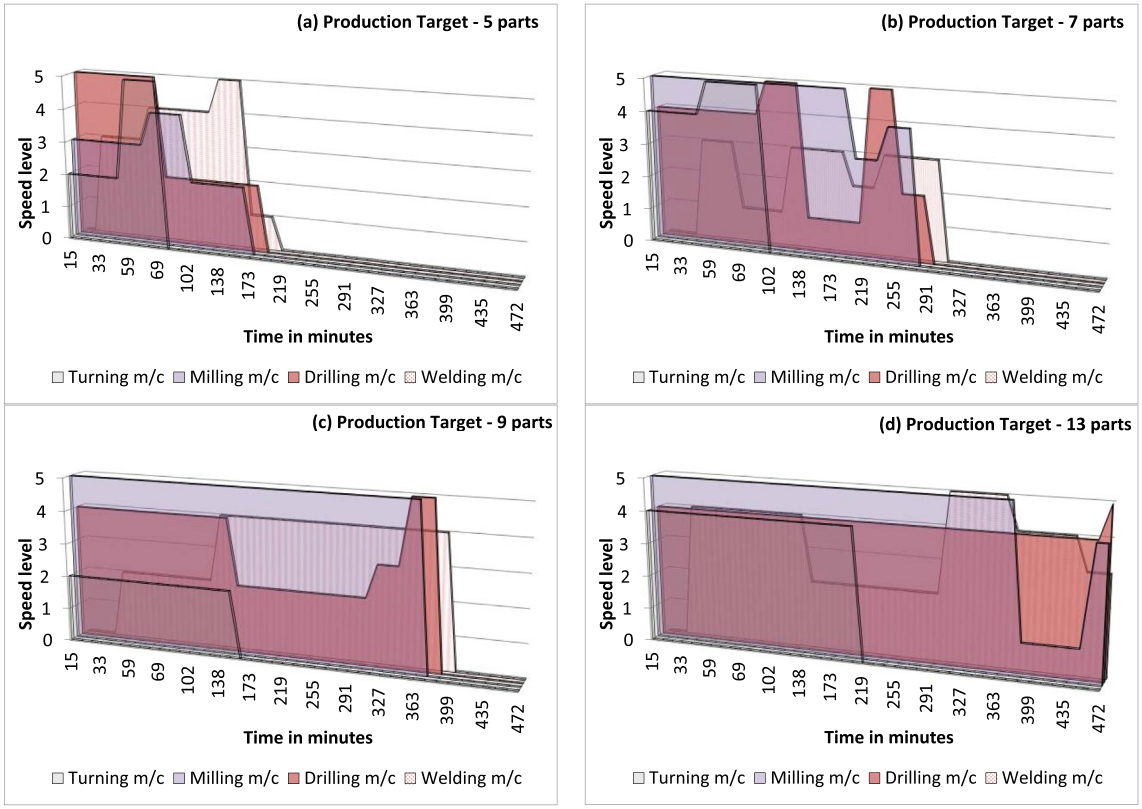


Fig. 7. Speeds change through the shift at different machines for target production (a) 5 parts per shift, (b) 7 parts per shift, and (c) 9 parts per shift (for the speed levels, refer to Tables 1 and 2).

change through the shift as a function of the machine. The speed levels in Fig. 7 refer to the speeds mentioned in Tables 1 and 2, and the milling and drilling are carried out on the same vertical machining center.

For smaller production targets, such as five parts per shift, as shown in Fig. 7(a), all of the machines run at moderate speeds for most of the time. As the production target increases to seven, as shown in Fig. 7(b), the milling must almost always be carried out at higher speeds to meet the target. It is to be noted that the milling is the most time-consuming operation in the entire part production. However, the drilling and the welding operation run at lower speeds to counterbalance the higher-speed milling operation. Further increment in the production target to 9 and 13 parts, as shown in Fig. 7(c) and (d), forces the milling to run at maximum speed all of the time. Moreover, the drilling and welding operations also need to run at higher speeds. The foregoing speed changes imposed by the production target affect the energy consumption, cost, and cycle time. Fig. 8 shows the energy consumption per part, electricity cost per part, and cycle time per part under the schedules econologically optimized for different production targets (parts per shift).

It can be observed that the parts can be produced at a steady rate of energy consumption irrespective of the production target. As expected, for smaller production targets, the time per part is greater than the cycle times associated with larger production targets. The speeds under optimum schedules are such that the excess electrical load because of the higher operating speed due to the higher production rate is compensated by a shorter operational time, resulting in a consistent rate of energy consumption. However, with smaller production targets, a smaller electricity cost can be achieved. This is because most of the production can be completed within the off-peak and part-peak periods. In addition, the speeds are lower for fewer parts, as observed previously. This in turn avoids higher peak loads. Such outcomes are important for allowing a manufacturer to understand the cost and time consequences of production decisions and to modify product prices and delivery schedules as appropriate.

It is imperative to note that the foregoing effects of scheduling are possible due to changes in the operating speeds of the machines. As observed earlier, the number of speed changes also affects the cost and energy consumption. This is obvious because schedules with fewer speed changes would have less flexibility to meet the optimization objectives. On the other hand, the maximum number of speed changes is limited by the number of produced parts because our model assumes that speed changes can take place only after the completion of operations associated with a subpart on a given machine. In the next section, the interrelation between the production target and number speed changes under the optimum conditions is discussed.

4.3. Number of speed changes

The number of speed changes allowed per shift is considered to be a decision variable and allowed to take a value per the requirements of the schedule optimization. However, the actual number of speed changes, i.e., the number of speed changes in the near-optimal schedule, is usually less than the designated value for two reasons. First, the target is achieved earlier than the shift period; hence, not all of the designated speed changes are utilized. Second, the actual speed change occurs later than expected due to incomplete operation at a machine at the designated time. Fig. 9 shows the actual number of speed changes for various production targets.

It can be observed that the number of speed changes required to produce a part is considerably lower for higher production targets. Due to the lower production time per part with a higher production target, more parts are completed between two consecutive speed changes. This in turn reduces the average number of speed changes required per part produced. This is an important outcome. Although we have not considered any time or cost associated with the speed change in the present work, due consideration of such cost will decrease the difference between the total operational cost of a part produced under the higher-production-target regime and lower-production-target regime. Many more studies can be conducted to develop a more comprehensive understanding of the econological scheduling optimization, as discussed in the concluding section.

5. Summary and conclusions

In the present paper, we have presented a case of scheduling optimization for a discrete manufacturing system running under a time-varying electricity tariff. The proposed model has the three following novel capabilities.

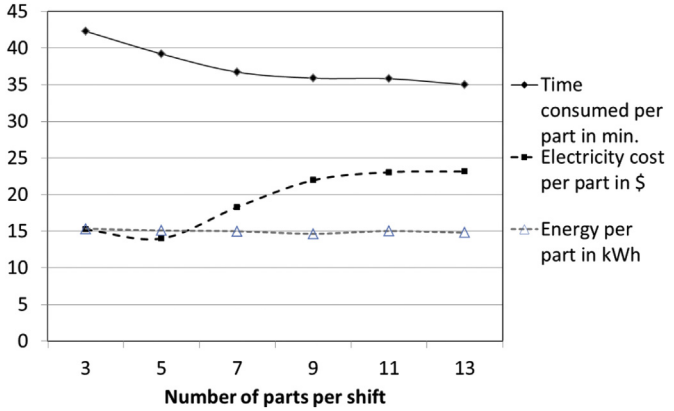


Fig. 8. Interrelation between production target, cost, and energy.

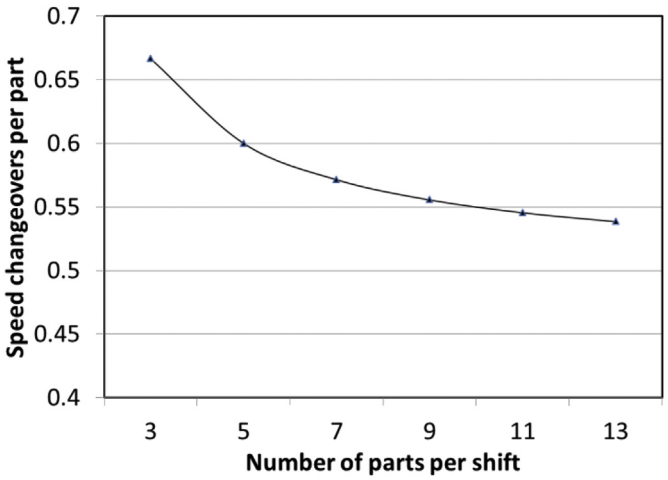


Fig. 9. Interrelation between production target and number of speed changes.

1. Conventional scheduling optimization in discrete manufacturing systems is limited to the scheduling of similar machines. The proposed formulation is demonstrated with a multiple-machine system comprised of machining and welding operations. This is characteristically different from the identical machine systems most often used in the scheduling optimization problems.
2. The model offers a combination of scheduling, discrete event simulation, and multi-criterion meta-heuristic optimization aimed at the simultaneous optimization of the electricity cost and energy consumption. The operational speed of the machines and the number of times the speed can change in a shift are considered as decision variables and allowed to change and affect the peak load, energy consumption, and electricity cost.
3. Unlike the traditional approach of using a scalar sum to address multi-criterion optimization problems, the present formulation uses aggregate indexing, which is observed to be more effective in handling conflicting objectives, as in the present case. Moreover, such an index prevents the formulation from being time-prohibitive and possesses the capacity to manage increasing numbers of objectives without computational issues.

A comparison between econological, economical, and ecological approaches is presented. The interrelationships between the production target, speed change, energy consumption, and electricity cost are discussed. Although the meta-heuristic optimization provided only near-optimal solutions, they are authoritative enough to portray the underlying dynamics of scheduling under a time-varying electrical tariff and provide strategic inputs to enable the manufacturing system to be more eco-friendly without substantial addition to the electricity cost. It is established that steady energy consumption per part can be achieved under the time-of-use tariff, even with varying production targets, by judiciously defining the objectives and optimizing the schedule. With a smaller production target, the electricity cost per part can be kept low; however, larger production targets require extra cost to maintain the eco-friendliness of the system. Nevertheless, the larger production targets can be competitive, as the non-productive costs are distributed.

The presented formulation has the potential for expansion to the industrial scale with additional objectives. The objectives may be more realistic and conflicting. It would be interesting to address the scheduling of an actual manufacturing enterprise comprised of a variety of operations with more complex interdependences. Future research may consider the balancing of production lines. The failure of a machine or maintenance may also be considered. Hybrid production systems, where both discrete and continuous or semi-continuous operations (e.g., the painting of body parts) take place, such as automobile production systems, may be quite challenging. The ‘basic’ and ‘idle’ components of power consumption associated with the utilities may also be addressed. Computational issues arising due to such complexity may require more extensive research. The application of the proposed formulation may also be attempted with other time-varying electricity tariffs expected to be used in the future by the power-supply companies, such as real-time pricing.

Appendix 1. Work flow and processing time calculation

a) Turning operation for cylindrical pins (A and B).

	Activity number (j)	Activity	D (mm)	Operational speed mm/min	Time in second	Power in Kw
Pin A	1	Loading			10	1.5
	2	Tool positioning			3.8	4.5
	3	Turning of entire length of 280 mm	25	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	4	Tool positioning			1.7	4.5
	5	Step turning (Right) pass 1	23	s_{1k}		$u \cdot f_r \cdot s_{1k} \cdot d$
	6	Tool positioning			1.2	4.5
	7	Step turning pass (Right) pass 2	21	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	8	Tool positioning			0.9	4.5
	9	Step turning (Left) pass 1	23	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	10	Tool positioning (b)			1.2	4.5
	11	Step turning (Left) pass 2	21	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	12	Tool change			6	4.5
	13	Parting			3	9
Pin A	14–26	Repetition of 1–13				
Pin B	27	Loading			10	1.5
	28	Tool positioning			3.8	4.5
	29	Turning of entire length of 280 mm	25	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	30	Tool positioning			1.7	4.5
	31	Step turning (Right) pass 1	23	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	32	Tool positioning			1.2	4.5
	33	Step turning pass (Right) pass 2	21	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	34	Tool positioning			0.9	4.5
	35	Step turning (Left) pass 1	23	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	36	Tool positioning (b)			1.2	4.5
	37	Step turning (Left) pass 2	21	s_{1k}	$\frac{\pi LD}{f_r s_{1k}}$	$u \cdot f_r \cdot s_{1k} \cdot d$
	38	Tool change			6	4.5
	39	Parting			3	9
Pin B	40–52		Repetition of 27–39			
Pin B	53–65		Repetition of 27–39			

b) Milling operation for plates (vertical and base).

Activity number	Activity	Operational speed mm/min	Time in second	Power in Kw
1	Loading		75	2
2	Tool positioning		3.2	7
3	Facing plate 1 − top	s_{2k}	$\frac{\pi.D_c(L+D_c)}{n_e f_t \cdot s_{2k}}$	$\frac{u \cdot W \cdot f_t \cdot s_{2k} \cdot d}{\pi}$
4	Tool home		3.2	7
5	Plate reversal		75	3
6	Facing plate 1 − bottom	s_{2k}	$\frac{\pi.D_c(L+D_c)}{n_e f_t \cdot s_{2k}}$	$\frac{u \cdot W \cdot f_t \cdot s_{2k} \cdot d}{\pi}$
7	Tool home		3.2	7
8	Unloading plate 1		150	3
9	Loading		75	3
10	Tool positioning		3.2	7
11	Facing plate 2 − top	s_{2k}	$\frac{\pi.D_c(L+D_c)}{n_e f_t \cdot s_{2k}}$	$\frac{u \cdot W \cdot f_t \cdot s_{2k} \cdot d}{\pi}$
12	Tool home		3.2	7
13	Plate reversal		75	3
14	Facing plate 2 − bottom	s_{2k}	$\frac{\pi.D_c(L+D_c)}{n_e f_t \cdot s_{2k}}$	$\frac{u \cdot W \cdot f_t \cdot s_{2k} \cdot d}{\pi}$
15	Tool home		3.2	7
16	Unclamping and clamping loading both plate		100	2
17	Tool change		7	2
18	Tool positioning		3.2	7
19	Drill hole one	s_{3k}	$\frac{L\pi D_c}{n_e f_t s_{3k}}$	$\frac{u \cdot n_e f_t \cdot s_{3k} \cdot D_c}{4}$
20	Tool positioning		6	7
21	Drill hole two	s_{3k}	$\frac{L\pi D_c}{n_e f_t s_{3k}}$	$\frac{u \cdot n_e f_t \cdot s_{3k} \cdot D_c}{4}$
22	Tool positioning		6	7
23	Drill hole three	s_{3k}	$\frac{L\pi D_c}{n_e f_t s_{3k}}$	$\frac{u \cdot n_e f_t \cdot s_{3k} \cdot D_c}{4}$
24	Tool positioning		6	7
25	Drill hole four	s_{3k}	$\frac{L\pi D_c}{n_e f_t s_{3k}}$	$\frac{u \cdot n_e f_t \cdot s_{3k} \cdot D_c}{4}$
26	Tool positioning		6	7
27	Drill hole five	s_{3k}	$\frac{L\pi D_c}{n_e f_t s_{3k}}$	$\frac{u \cdot n_e f_t \cdot s_{3k} \cdot D_c}{4}$
28	Tool positioning		3.2	7
29	Unloading + assembly		300	2

c) Welding operation.

Activity number	Activity	Operational speed mm/min	Time in second	Power in Kw
1	Loading		40	1
2	Torch movement		3	2
3	Tack-weld 1 (Left and right plate)*		3.75	$VQ + P_t$
4	Torch movement		3.75	2
5	Tack-weld 2 (Left and right plate)		3.75	$VQ + P_t$
6	Torch movement		3.75	2
7	Tack-weld 3 (Left and right plate)		7.5	$VQ + P_t$
8	Torch movement		3.75	2
9	Tack-weld 4 (Left and right plate)		3.75	$VQ + P_t$
10	Torch movement		7.5	2
11	Welding time (Pass 1)&,#	s_{4k}	$\frac{L}{s_{4k}}$	$VQ + P_t$
12	Inter-pass time		600	
13	Welding time (Pass 2)	s_{4k}	$\frac{L}{s_{4k}}$	$VQ + P_t$
14	Inter-pass time		600	
15	Welding time (Pass 3)	s_{4k}	$\frac{L}{s_{4k}}$	$VQ + P_t$
16	Unloading		40	1

*The tack welds of 20 mm length are made at constant welding speed of 320 mm/min.
&The welding set-up consists of two robotic welders that simultaneously welds the left and right vertical plates with the base plate.
#Three pass weld.

Appendix 2

(b) Discrete Event Simulation: The chart shown in Fig. 10 represents discrete event simulation for a one machine set-up for running under a TOU tariff. The algorithm gives electrical load (power) as a function of incremental time (Δt) for a duration of time t_{max}. The time−power curve is used in calculate maximum power, electrical consumption (area under the time−power curve). The activity time $T(a)$ is obtained from a typical work-flow given in Appendix 1.

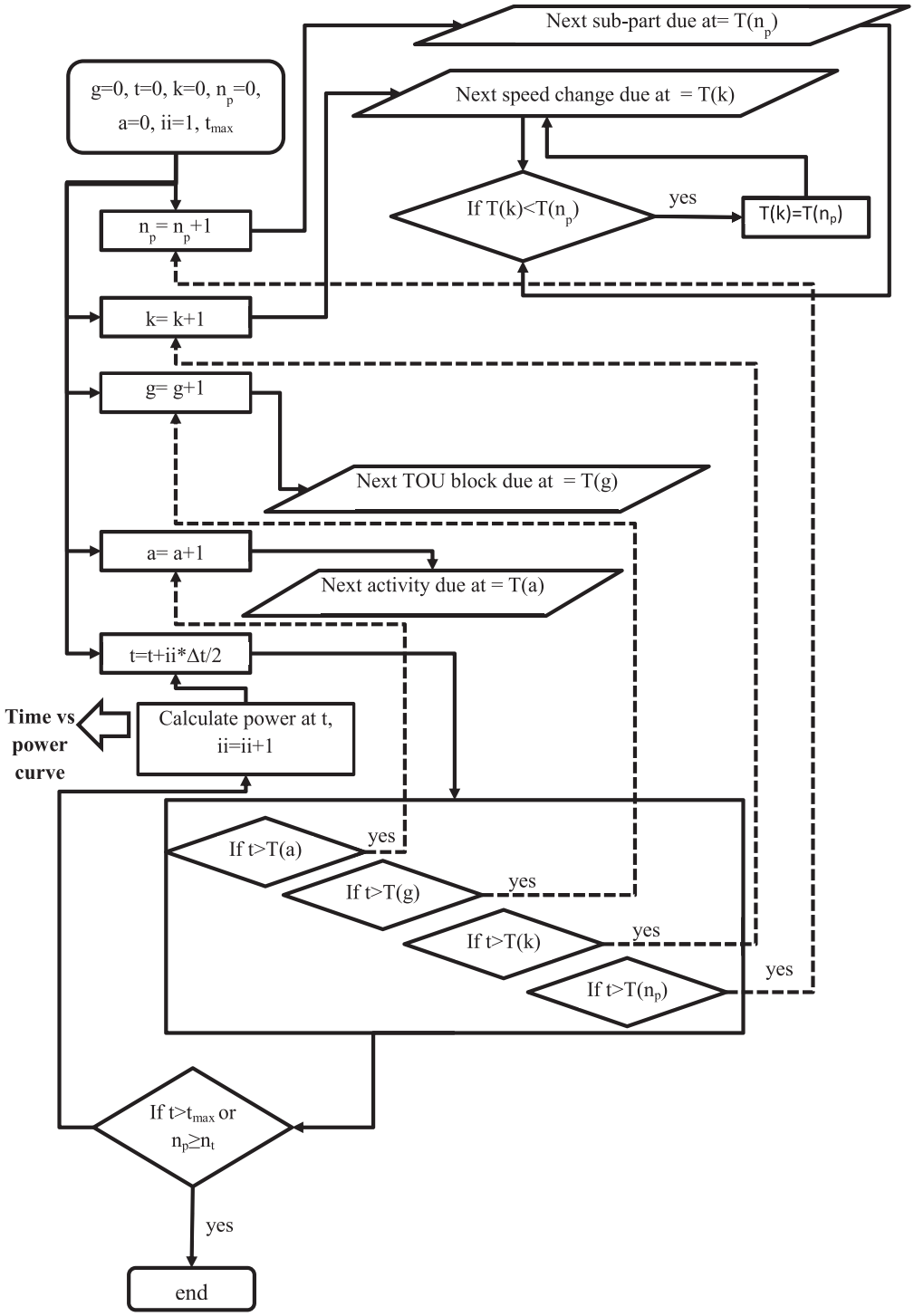


Fig. 10. Discrete Event Simulation.

References

Alain, Hait, Christian, Artigues, 2011. On electrical load tracking scheduling for a steel plant. Comput. Chem. Eng. 35, 3044–3047.

Albadi, M.H., El-Saadany, E.F., 2008. A summary of demand response in electricity markets. Electr. Power Syst. Res. 78 (11), 1989–1996.

Allahverdi, A., Al-Anzi, F.S., 2009. The two-stage assembly scheduling problem to minimize total completion time with setup times. Comput. Oper. Res. 36 (10), 2740–2747.

Armentano, V.A., Arroyo, J.E.C., 2004. An application of a multi objective tabu search algorithm to a bi-criteria flow shop problem. J. Meta-heuristics 10, 463–481.

Ashok, S., 2006. Peak-load management in steel plants. Appl. Energy 83, 413–424.

Basu, S., Sutherland, J.W., 1999. Multi-objective decision making for environmental responsible manufacturing. In: Proceedings of 6th CIRP International Seminar on Life Cycle Engineering, pp. 321–331.

Blum, C., 2005. Beam-ACO hybridizing ant colony optimization with beam search: an application to open shop scheduling. Comput. Oper. Res. 32 (6), 1565–1591.

Brandimarte, P., 1993. Routing and scheduling in a flexible job shop by tabu search. Ann. Oper. Res. 41 (3), 157–183.
Brucker, P., Schlie, R., 1990. Job-shop scheduling with multi-purpose machines. Computing 45 (4), 369–375.
Chang, P.C., Chen, S.H., Liu, C.H., 2007. Sub-population genetic algorithm with mining gene structures for multi-objective flow shop scheduling problems. Expert Syst. Appl. 33, 762–771.
Chen, Ruey Maw, 2011. Particle swarm optimization with justification and designed mechanisms for resource-constrained project scheduling problem. Expert Syst. Appl. 38, 7102–7111.
Dahmus, J.B., Gutowski, T.C., 2004. An Environmental Analysis of Machining, ASME International Mechanical Engineering Congress and RD&D Expo. California USA, Anaheim.
Dietrich, B.L., 1991. A taxonomy of discrete manufacturing systems. Oper. Res. 39 (6), 886–902.
Duflou, R.J., Sutherland, J.W., Dornfeld, David, Christoph, Herrman, Jack, Jeswiet, Sami, Kara, Michael, Hauschild, Karel, Kellens, 2012. Towards energy and resource efficient manufacturing: a processes and systems approach. CIRP Ann. Manuf. Technol. 61, 587–609.
Fadi, Shrouf, Joaquin, Ordieres-Meré, Alvaro, García-Sánchez, Miguel, Ortega-Mier, 2014. Optimizing the production scheduling of a single machine to minimize total energy consumption costs. J. Clean. Prod. 67, 197–207.
Fang, Kan, Nelson, Uhan, Fu, Zhao, Sutherland, John W., 2011. A new approach to scheduling in manufacturing for power consumption and carbon footprint reduction. J. Manuf. Syst. 3, 234–240.
Gonzalez, T., Sahni, S., 1976. Open shop scheduling to minimize finish time. J. ACM 23 (4), 665–679.
Gonçalves, J.F., de Magalhães Mendes, J.J., Resende, M.G., 2005. A hybrid genetic algorithm for the job shop scheduling problem. Eur. J. Oper. Res. 167 (1), 77–95.
He, Y.A.N., Liu, F.E.I., Shi, J., 2008. A framework of scheduling models in machining workshop for green manufacturing. J. Adv. Manuf. Syst. 7 (02), 319–322.
He, Y., Liu, B., Zhang, X., Gao, H., Liu, X., 2012. A modeling method of task-oriented energy consumption for machining manufacturing system. J. Clean. Prod. 23 (1), 167–174.
Herrmann, C., Thiede, S., Kara, S., Hesselbach, J., 2011. Energy oriented simulation of manufacturing systems – concept and application. CIRP Ann. Manuf. Technol. 60, 45–48.
Jackson, Paul, Wallace, David, Harold, Esther, Edgerton, E., Kegg, Richard, 1997. An analytical method for integrating environmental and traditional design considerations. CIRP Ann. Manuf. Technol. 46, 355–360.
Jacopo, Torriti, 2012. Price-based demand side management, assessing the impacts of time-of-use tariffs on residential electricity demand and peak shifting in Northern Italy. Energy 44, 576–583.
Kristian, Nolde, Manfred, Morari, 2010. Electrical load tracking scheduling of a steel plant. Comput. Chem. Eng. 34, 1899–1903.
Lee, C.Y., Cheng, T.C.E., Lin, B.M.T., 1993. Minimizing the makespan in the 3-machine assembly-type flowshop scheduling problem. Manag. Sci. 39 (5), 616–625.
Leung, C.W., Wong, T.N., Maka, K.L., Fung, R.Y.K., 2010. Integrated process planning and scheduling by an agent-based ant colony optimization. Comput. Ind. Eng. 59, 166–180.
Marcus, Frantzen, Amos, Ng, H.C., Moore, Philip Marcus, 2011. A simulation-based scheduling system for real-time optimization and decision making support. Robot. Computer-Integrated Manuf. 27, 696–705.
María, José Arbiza, Anna, Bonfill, Gonzalo, Guillén, Fernando, D. Mele, Antonio, Espuña, Luis, Puigjaner, 2008. Metameta-heuristic multiobjective optimisation approach for the scheduling of multiproduct batch chemical plants. J. Clean. Prod. 16 (2), 233–244.
Mezmaz, M., Melab, N., Kessaci, Y., Lee, Y.C., Talbi, E.G., Zomaya, A.Y., Tuyttens, D., 2011. A parallel bi-objective hybrid metaheuristic for energy-aware scheduling for cloud computing systems. J. Parallel Distributed Comput. 71 (11), 1497–1508.
Mouzon, G., Yildirim, M.B., 2008. A framework to minimise total energy consumption and total tardiness on a single machine. Int. J. Sustain. Eng. 1, 105–116.
Munoz, A.A., Sheng, P., 1995. An analytical approach for determining the environmental impact of machining processes. J. Mater. Process. Technol. 53, 736–758.
Natasha, Gr, Vaklieva-Bancheva, Kirilova, Elisaveta G., 2010. Cleaner manufacture of multipurpose batch chemical and biochemical plants. Scheduling and optimal choice of production recipes. J. Clean. Prod. 18 (13), 1300–1310.
Osman, I.H., Potts, C.N., 1989. Simulated annealing for permutation flow-shop scheduling. Omega 17 (6), 551–557.
Pezzella, F., Morganti, G., Ciaschetti, G., 2008. A genetic algorithm for the flexible job-shop scheduling problem. Comput. Oper. Res. 35 (10), 3202–3212.
Potts, C.N., Sevast'Janov, S.V., Strusevich, V.A., Van Wassenhove, L.N., Zwaneveld, C.M., 1995. The two-stage assembly scheduling problem: complexity and approximation. Oper. Res. 43 (2), 346–355.
Prabhat, K. Swamee, Aditya, Tyagi, 2000. Describing water quality with aggregate Index. J. Environ. Eng. 126, 451–455.
Róbert, Adonyi, Gábor, Biros, Tibor, Holczinger, Ferenc, Friedler, 2008. Effective scheduling of a large-scale paint production system. J. Clean. Prod. 16 (2), 225–232.
Sayadi, M., Ramezanian, R., Ghaffari-Nasab, N., 2010. A discrete firefly metaheuristic with local search for makespan minimization in permutation flow shop scheduling problems. Int. J. Ind. Eng. Comput. 1 (1), 1–10.
Schlosser, R., Klocke, F., Lung, D., 2011. Sustainability in manufacturing –energy consumption of cutting processes, advances in sustainable manufacturing. In: Proceedings of the 8th Global Conference on Sustainable Manufacturing, Heidelberg.
Sheng, P., Bennetz, D., Thurwachterl, S., 1998. Environmental-based systems planning for machining. CIRP Ann. Manuf. Technol. 47, 409–413.
Shrouf, F., Ordieres-Meré, J., García-Sánchez, A., Ortega-Mier, M., 2014. Optimizing the production scheduling of a single machine to minimize total energy consumption costs. J. Clean. Prod. 67, 197–207.
Subai, C., Baptiste, N.E., 2006. Scheduling issues for environmentally responsible manufacturing, the case of hoist scheduling in an electroplating line. Int. J. Prod. Eng. 99, 74–87.
Sun, Zeyi, Li, Lin, 2014. Potential capability estimation for real time electricity demand response of sustainable manufacturing systems using Markov Decision Process. J. Clean. Prod. 65, 184–193.
Swaminathan, V., Chakrabarty, K., 2003. Energy-conscious, deterministic I/O device scheduling in hard real-time systems. IEEE Trans. Computer-Aided Des. Integr. Circuits Syst. 22, 847–858.
Tavakkoli-Moghaddam, R., Rahimi-Vahed, A., Mirzaei, A.H., 2007. A hybrid multi-objective immune algorithm for a flow shop scheduling problem with bi-objectives: weighted mean completion time and weighted mean tardiness. Inf. Sci. 177 (22), 5072–5090.
Vallada, E., Ruiz, R., 2009. Cooperative metaheuristics for the permutation flowshop scheduling problem. Eur. J. Oper. Res. 193 (2), 365–376.
Varadharajan, T.K., Rajendran, C., 2005. A multi-objective simulated- annealing algorithm for scheduling in flow shops to minimize the makespan and total flowtime of jobs. Eur. J. Oper. Res. 167, 772–795.
Wang, J., Li, J., Huang, N., 2009. Optimal scheduling to achieve energy reduction in auto-motive paint shops. In: Proceedings of 2009 ASME Manufacturing Science and Engineering Conference, West Lafayette, Indiana, USA.
Yin-Yann, Chen, Chen-Yang, Cheng, Li-Chih, Wang, Tzu-Li, Chen, 2013. A hybrid approach based on the variable neighbourhood search and particle swarm optimization for parallel machine scheduling problems – a case study for solar cell industry. Int. J. Prod. Econ. 141 (1), 66–78.
Yoon, H.S., Moon, J.S., Pham, M.Q., Lee, G.B., Ahn, S.H., 2013. Control of machining parameters for energy and cost savings in micro-scale drilling of PCBs. J. Clean. Prod. 54, 41–48.
Zobolas, G.I., Tarantilis, C.D., Loannou, G., 2009. Minimizing makespan in permutation flow shop scheduling problems using a hybrid metaheuristic algorithm. Comput. Oper. Res. 36 (4), 1249–1267.